What is JSON Web Token?

An open standard (RFC 7519) called JSON Web Token (JWT) specifies a small, self-contained method for safely sending data as a JSON object between parties. Because this information is digitally signed, it can be confirmed and trusted. JWTs can be signed using a public/private key pair using RSA or ECDSA, or with a secret (using the HMAC technique).

When should we use JSON Web Tokens?

- Authorization: This is the most typical JWT use case. After logging in, the JWT will be included in all subsequent requests, enabling the user to access resources, services, and routes that are allowed with that token. These days, Single Sign On is a feature that makes extensive use of JWT due to its low overhead and ease of use across several domains.

- Information Exchange: Secure information transmission between parties is made possible using JSON Web Tokens. You can be certain that the senders are who they claim to be since JWTs can be signed, for instance, using public/private key pairs. You may also confirm that the content hasn't been altered because the signature is computed using both the header and the payload.

JWT Structure

JWT is a string made up of three parts separated by dots (.)

Header

Contains metadata about the token

```
"alg": "HS256",
"typ": "JWT"
```

→

Payload

Contains the claims (data)

```
"sub": "1234567890",
"name": "John Doe",
"role": "admin",
"iat": 1716190000,
"exp": 1716193600
```

→

Signature

Used to verify that the token is authentic

Authenticity check

The signature confirms the token has not been altered.